

Local Credit, Global Settlement

Enhancing Trust-Based Credit Creation with On-Chain Reserve Contracts

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Ergo

8th CCCS Conference — RAMICS

Plurality of Social Currencies: solidarity economy, municipal and community currencies for inclusive and sustainable futures

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Offchain Payments Today: No Monetary Expansion

Existing Bitcoin Layer-2 Systems

System	Mechanism	Collateral Requirement
Lightning Network	Bidirectional payment channels	100% locked on-chain
Cashu	Chaumian blind signatures (e-cash)	100% mint reserves
Fedimint	Federated threshold custody	100% federation backing

“ All three require full backing of off-chain liabilities with on-chain assets. ”

The Problem: No Credit Expansion

- Supply is capped by locked collateral
- Cannot respond to growing economic activity

Contrast with Community Currencies

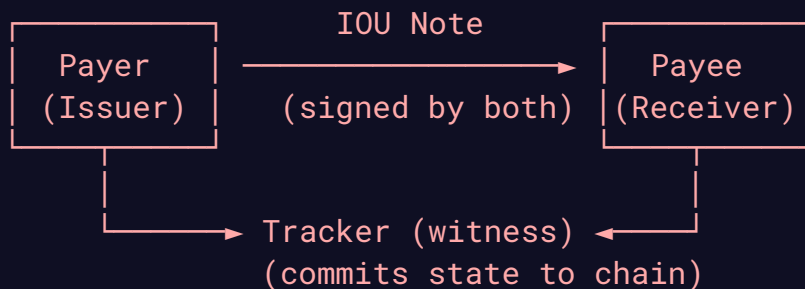
- Community currencies (LETS, Time Banks, WIR) thrive on **trust-based credit**
- Existing digital systems **discard this principle** by enforcing full collateralization
- **Trust within communities becomes an underutilized resource**

Basis: A Protocol to unify Community Credit and Blockchain Assets

“Enable elastic credit creation backed by trust, use on-chain reserves only when trust is insufficient”

How It Works

- **Local trust-based IOUs** circulate within communities
- **On-chain reserves** serve as optional backing
- Individual IOU note acceptance
- **Off-chain payments** — low fees, no blockchain bottleneck
- **Tracker service** coordinates debt state transparently



Note Creation: Issuing Community Credit

1. **Payer** (e.g., a community member) creates an IOU note:
 - Payer's **signature**
 - Payee's **public key**
 - **Debt amount**
 - **Timestamp**
 - Optional: **reserve contract** for extra backing
2. **Tracker** (community coordinator / federation) verifies and **signs**
 - Certifies inclusion in authenticated state
3. **Payee** accepts after verifying both signatures. **Acceptance is always voluntary**

Note Redemption: From Trust to Settlement

Standard Redemption (Trust + Collateral)

- Payee presents the **witnessed IOU** to the **reserve contract**
- Contract verifies payer's signature, tracker's signature, debt amount
- Collateral is **released to payee**

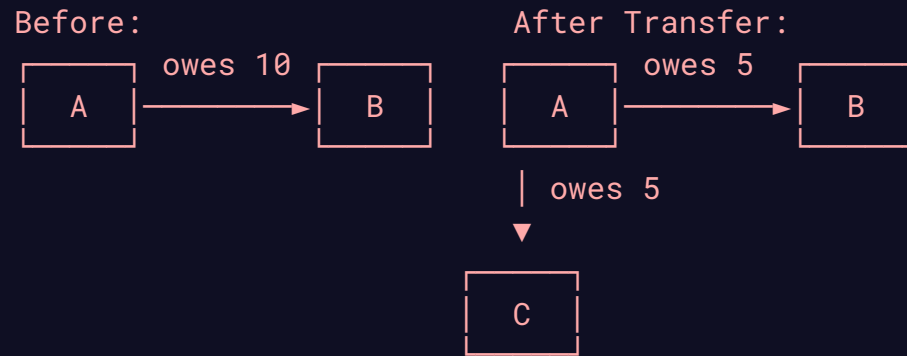
Emergency Exit (Tracker Unavailable)

- After **3 days (2,160 blocks)**, tracker signature becomes **optional**
- Redeem using payer's signature + last committed tracker state
- **Replay attacks prevented** by timestamp verification

“This protects communities against tracker failure or censorship — funds are never locked indefinitely.”

Debt Transfer: Circulating Credit Within Communities

Triangular Trade (No Blockchain Needed)



- **B** requests **A** to co-sign:
 - Reduced debt $A \rightarrow B$
 - New debt $A \rightarrow C$
- Tracker witnesses both; old note replaced
- **A now owes C directly** — credit circulates!

Tracker State: Transparent Debt Registry

AVL Tree of Debt Relationships

$\text{hash}(\text{payerKey} || \text{payeeKey}) \rightarrow (\text{totalDebt}, \text{timestamp})$

- Cryptographically strong hash function
- Efficient lookups and **proof generation**
- Root digest **committed to blockchain periodically**
- Enables **emergency redemption** and **public verification**

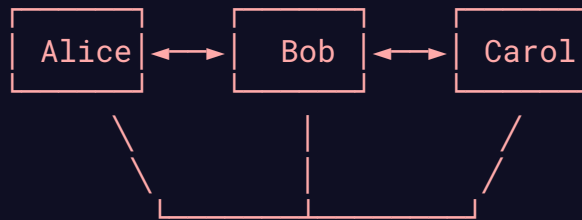
Governance Options

Design	Trust Model	Best For
Single Server	Trusted operator	Small communities
Federation (MuSig2)	Threshold of signers	Municipal currencies
Sidechain (PoW)	Permissionless consensus	Regional / multi-community

Scenario 1: Rural Community Trading

Disconnected Village with Occasional Connectivity

Village (no direct Internet)



Local Tracker
(syncs via satellite/cellular)

- Trade on **credit** within community over **Bluetooth / LoRa**
- Tracker commits to blockchain **when Internet available**
- Redemption possible via **slow links (SMS, email)**
- **No forced collateralization** — trust is enough locally

Scenario 2: Municipal Currency with Gradual Backing

A City-Scale Complementary Currency

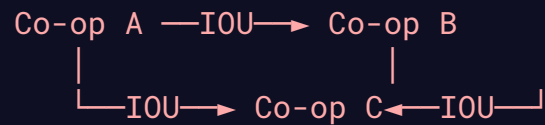
1. **Phase 1 — Pure Trust:** Local businesses issue IOUs to each other
 - No collateral required
 - Tracker (city federation) witnesses transactions
2. **Phase 2 — Selective Backing:** As trade expands to outsiders
 - Businesses optionally create **on-chain reserves**
 - Reserves back IOUs for **inter-city / low-trust** transactions
3. **Phase 3 — Hybrid Economy:** Local trust dominates internally, reserves handle external settlement

“**Monetary elasticity:** supply expands with community trust, contracts when trust is insufficient.”

Scenario 3: Solidarity Economy Network

Cooperatives and Mutual Credit

- **Worker cooperatives** trade goods/services on credit
- **Mutual credit clearing** via debt transfer (triangular trade)
- **Tracker federation** run by cooperative assembly
- **Emergency exit** protects members against federation failure



Circular debt cleared via tracker
No blockchain fees for day-to-day trade

Security & Trust Minimization

Tracker Cannot Steal Funds

- Tracker only **certifies inclusion** in state
- **Issuer's signature always required** for redemption
- Tracker cannot unilaterally create or redeem notes

Anti-Censorship

- Pseudonymity allows picking new keys
- Reserve contract has **refund option** (long timeout)
- Future: blind signatures + zero-knowledge proofs

Emergency Exit

- After timeout: tracker signature **optional**
- Same message format: `key || totalDebt || timestamp`
- Reserve owner's signature still required

Why Blockchain? Why Ergo?

Role of On-Chain Reserves

- **Not** for day-to-day transactions
- **Settlement layer** for low-trust or cross-community trades
- **Emergency anchor** if tracker fails
- **Limited blockchain use** — only where trust is truly insufficient

Why Ergo?

- **UTXO model**: perfect for tokenized collateral
- **Smart contracts**: expressive redemption logic
- **Low fees**: viable for small settlements
- **PoW security**: no trusted validators
- **NiPoPoWs / SPV**: lightweight sync for intermittent connectivity

Relation to Conference Themes

Theme	How Basis Addresses It
Digitalization & technologies	Blockchain-backed IOUs, mesh network support, digital sovereignty
Territories & participation	Local credit for communities, voluntary acceptance, democratic tracker governance
Public policies & government relations	Municipal currency potential, no legal tender enforcement, individual choice
Environment & sustainability	Minimal energy use (off-chain payments), no mining for transactions
History & theories	Mutual credit tradition extended with cryptographic trust minimization

Contributions

1. **Basis protocol**: P2P credit creation with **optional** on-chain reserves
2. **Open-source implementations**: smart contracts + tracker server + clients
3. **Security analysis**: trust-minimization properties of tracker
4. **Real-world demos**: mesh network trading + solidarity economy scenarios

Resources

- Paper & code: github.com/ChainCashLabs/chaincash
- Whitepaper: [docs/conf/conf.pdf](#)

Thank You

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github.com/ChainCashLabs/chaincash

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